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(54) **COAXIAL SPEAKER**

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H04R 7/04	(2006.01)
H04R 7/18	(2006.01)
H04R 9/02	(2006.01)
H04R 9/04	(2006.01)
H04R 9/06	(2006.01)

(52) **U.S. Cl.**

CPC ***H04R 1/24*** (2013.01); ***H04R 1/025*** (2013.01); ***H04R 7/04*** (2013.01); ***H04R 7/18*** (2013.01); ***H04R 9/025*** (2013.01); ***H04R 9/045*** (2013.01); ***H04R 9/063*** (2013.01); *H04R 2400/11* (2013.01)

(58) **Field of Classification Search**

CPC . H04R 1/24; H04R 1/025; H04R 7/04; H04R 7/18; H04R 9/025; H04R 9/045; H04R 9/063; H04R 2400/11

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,772,466 A * 11/1973 Hossbach H04R 1/2834 381/396
6,343,128 B1 * 1/2002 Coffin H04R 9/06 381/186

(Continued)

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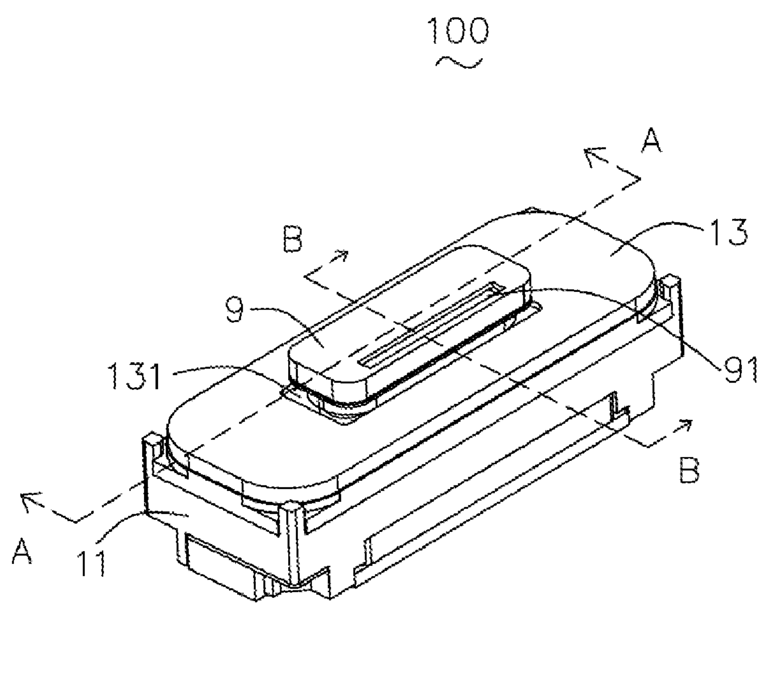
(57) **ABSTRACT**

The present invention provides a coaxial speaker including a fixing member, a magnetic circuit system, a voice coil, a first vibration system connected with the voice coil, and a second vibration system connected with the voice coil and arranged surrounding the first vibration system. The second vibration system locates coaxially with the first vibration system. The first vibration system includes a first diaphragm and a first support connecting the first diaphragm and the voice coil. The second vibration system includes a second diaphragm and a second support connecting the second diaphragm and the voice coil. The first diaphragm is used for generating treble sound. The second diaphragm is used for generating bass sound. Bass sound and treble sound form a full-range speaker, providing high-quality sound. Moreover, the assembly process is simplified by sharing voice coil, which improves assembly efficiency and reduces costs.

8 Claims, 4 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

6,714,656	B1 *	3/2004	Coffin	H04R 9/02 381/420
8,831,270	B1 *	9/2014	Dimitrov	H04R 1/30 381/401
10,631,096	B1 *	4/2020	Garcia Selva	H04R 9/06
2011/0176703	A1 *	7/2011	Horigome	H04R 1/2842 381/412
2014/0056465	A1 *	2/2014	Li	H04R 9/025 381/400
2014/0241566	A1 *	8/2014	Choi	H04R 9/043 381/400
2015/0237449	A1 *	8/2015	Papp	H04R 7/02 381/398
2020/0275189	A1 *	8/2020	Cho	H04R 7/16
2025/0008269	A1 *	1/2025	Zhang	H04R 9/045

* cited by examiner

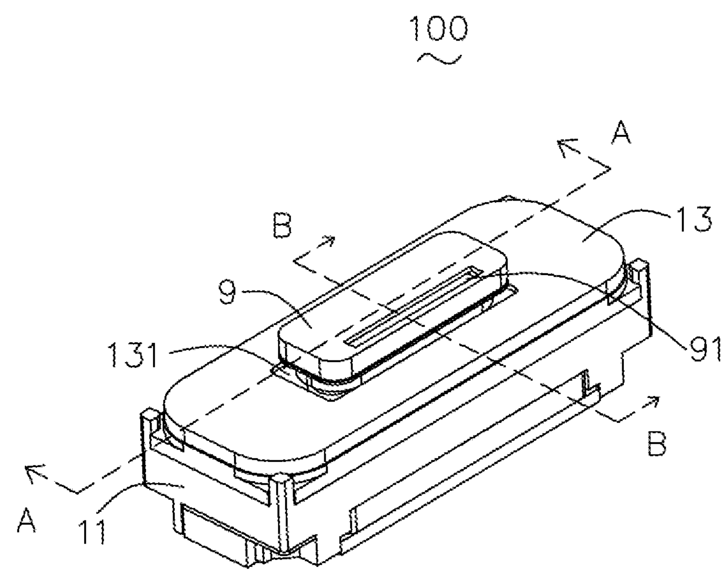


FIG. 1

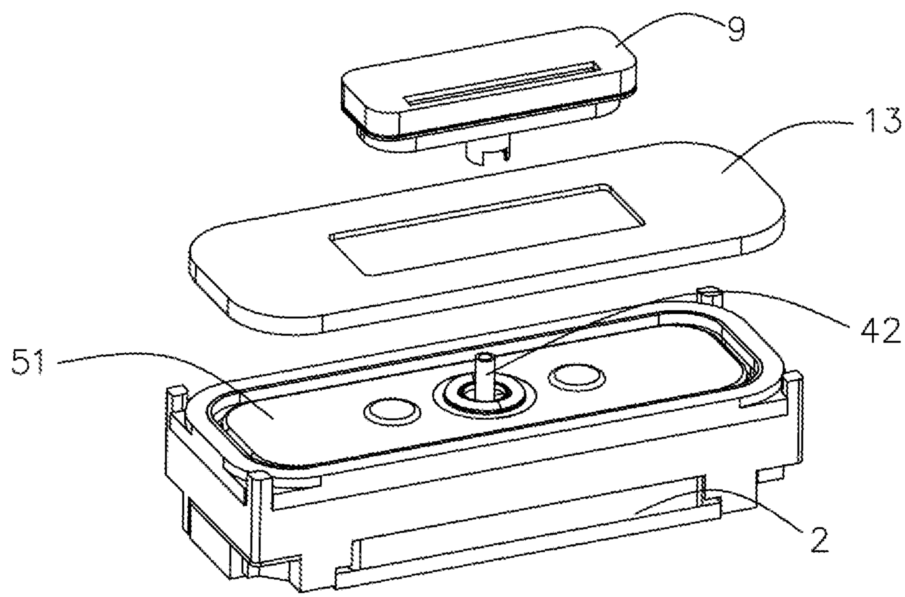


FIG. 2

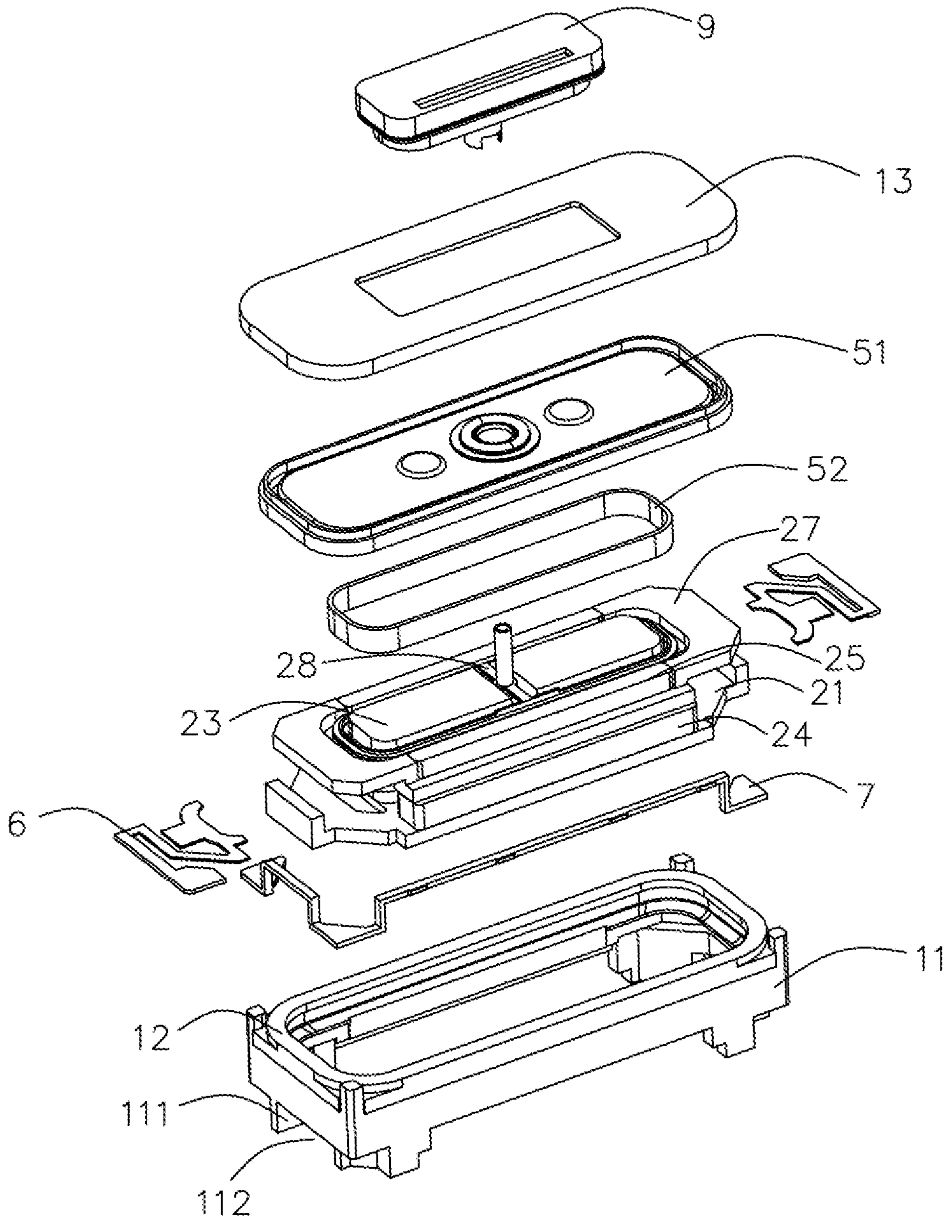


FIG. 3

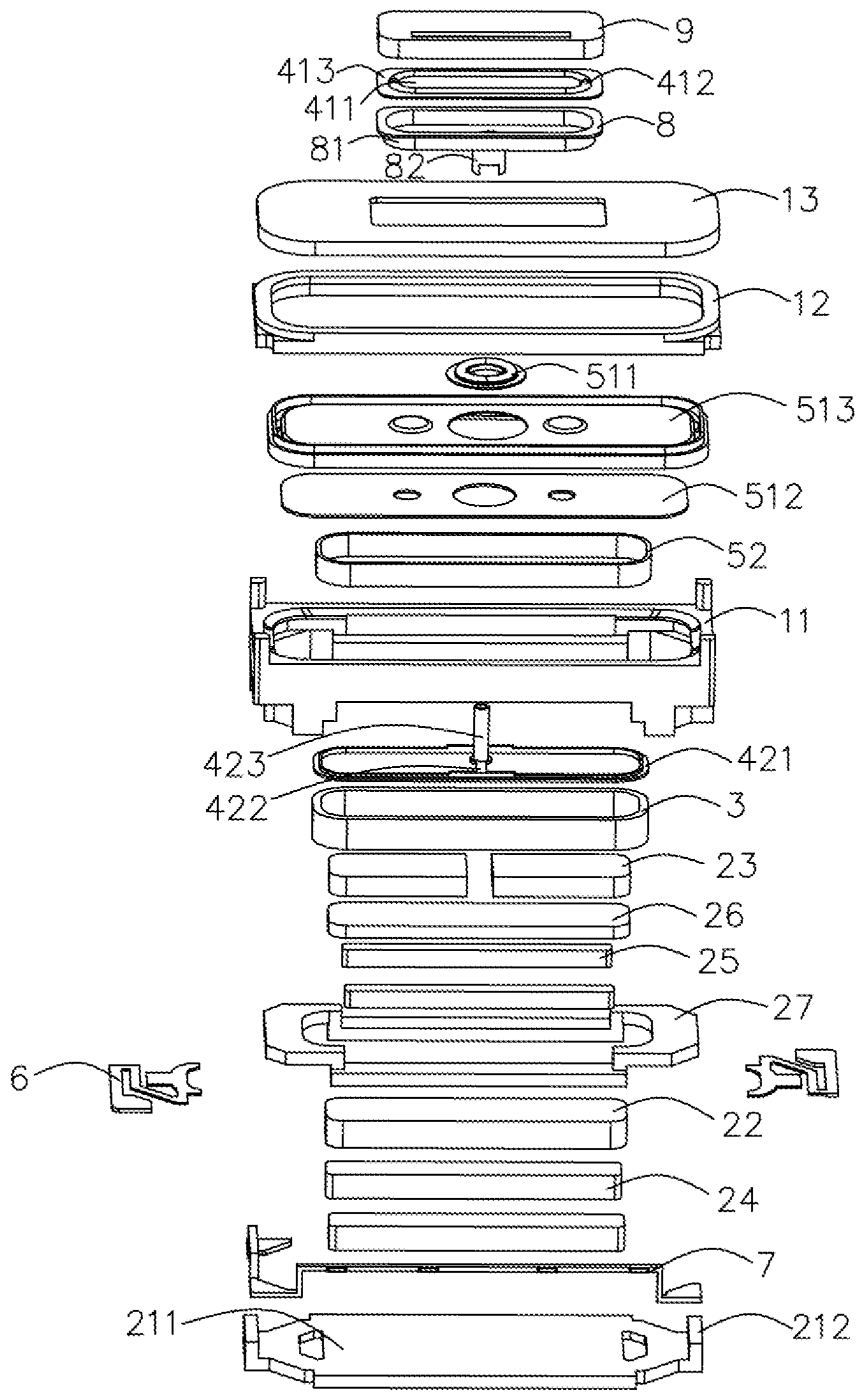

9
413
412
411
8
81
82
13
12
511
513
512
52
11
423
421
422
3
23
26
25
27
6
22
24
7
211
212

FIG. 4

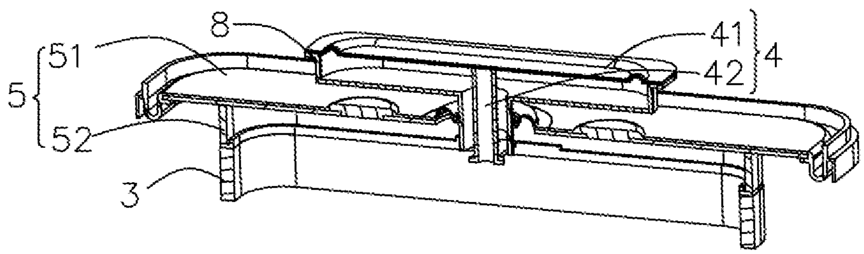


FIG. 5

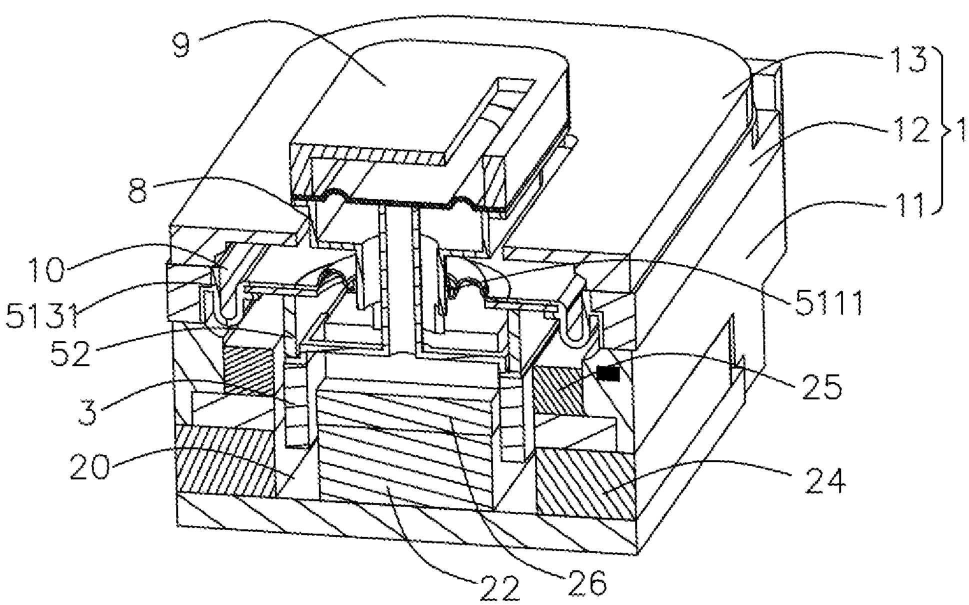


FIG. 6

COAXIAL SPEAKER

FIELD OF THE PRESENT INVENTION

The present invention relates to the field of electro-acoustic devices, and more particularly to a coaxial speaker.

DESCRIPTION OF RELATED ART

Coaxial speaker refers to the speaker with the integration of the treble unit and bass unit, and the treble unit and bass unit are set coaxially. The sound source of the treble unit and bass unit of the coaxial speaker can be located in front of the coaxial speaker, that is, the two speaker units sounding along a same direction. The sound source of the treble unit and bass unit of the coaxial speaker can also be located in front and back of the coaxial speaker respectively, that is, the two speaker units sounding along different directions.

In related art, the treble unit and bass unit of the coaxial speaker usually each includes an independent vibration system and an independent magnetic circuit system, and the structure of the coaxial speaker is complex. Because the coaxial speaker is consisted of the treble unit and the bass unit coaxially, the overall size of coaxial speaker is increased, which resulting in inconvenient installation and reduced the suitability of the product.

Therefore, it is desired to provide a new coaxial speaker which can overcome the above problems.

SUMMARY

In view of the above, the embodiments of the present invention provide a new coaxial speaker. By the present invention, the assembly structure of coaxial speaker is simple and the cost is reduced.

The present invention provides a coaxial speaker including a fixing member having a receiving room, a magnetic circuit system mounted with the fixing member and having a magnetic gap, a voice coil at least partially located into the magnetic gap, a first vibration system connected with the voice coil, and a second vibration system connected with the voice coil and arranged surrounding the first vibration system. The second vibration system locates coaxially with the first vibration system. The first vibration system includes a first diaphragm and a first support connecting the first diaphragm and the voice coil. The second vibration system includes a second diaphragm and a second support connecting the second diaphragm and the voice coil. The first diaphragm is used for generating treble sound. The second diaphragm is used for generating bass sound.

As an improvement, the second diaphragm is closer to the magnetic circuit system than the first diaphragm.

As an improvement, the first support includes a first fixing portion with an annular shape connecting to the voice coil, a connecting portion extending inwardly from the first fixing portion, and a second fixing portion extending from the connecting portion toward the first diaphragm and connected to the first diaphragm.

As an improvement, the second diaphragm includes a first suspension with an annular hollow shape, a first dome surrounding the first suspension, and a second suspension partially surrounding the first dome, the second fixing portion of the first support passing through the first suspension and connected to the first diaphragm.

As an improvement, the second support has an annular shape, one end of the second support connected with the first dome, and the other end of the second support connected to with the voice coil.

As an improvement, one end of the second support distal to the second diaphragm is connected to the first fixing portion of the first support so that the first fixing portion is sandwiched between the second support and the voice coil.

As an improvement, the coaxial speaker further includes a supporting member connecting the first diaphragm to the second diaphragm, the supporting member including a first portion located out of the receiving room of the fixing member and a second portion located in the receiving room, the first portion connected with the first diaphragm, the second portion connected with the second diaphragm, a size of the second portion smaller than the size of the first portion.

As an improvement, the coaxial speaker further includes a first housing covered above the first diaphragm, a first sound hole is provided on the first housing for sounding the treble sound generated by the first diaphragm.

As an improvement, the fixing member includes a frame secured to the magnetic circuit system, a supporting ring secured to the second diaphragm, and a second housing mounted with the supporting ring, the second housing covering the second diaphragm and spaced apart from the supporting member to form a second sound hole for sounding the bass sound generated by the second diaphragm.

As an improvement, the second portion of the supporting member extends to and fix with the magnetic circuit system.

BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the exemplary embodiments can be better understood with reference to the following drawing. The components in the drawing are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present invention. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. **1** is an illustrative isometric view of a coaxial speaker in accordance with one embodiment of the present invention.

FIG. **2** is a partially exploded view of the coaxial speaker of FIG. **1**.

FIG. **3** is another partially exploded view of the coaxial speaker of FIG. **1**.

FIG. **4** is an exploded view of the coaxial speaker of FIG. **1**.

FIG. **5** is an illustrative cross-sectional view of some members of the coaxial speaker taken along line A-A of FIG. **1**.

FIG. **6** is an illustrative cross-sectional view of the coaxial speaker taken along line B-B of FIG. **1**.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

The present invention will hereinafter be described in detail with reference to exemplary embodiments. To make the technical problems to be solved, technical solutions and beneficial effects of the present invention more apparent, the present invention is described in further detail together with the figures and the embodiments. It should be understood the specific embodiments described hereby is only to explain the disclosure, not intended to limit the disclosure.

Referring to the FIGS. **1-6**, the embodiment of the present invention provides a coaxial speaker **100**. The coaxial speaker **100** includes a fixing member **1** having a receiving room **10**, a magnetic circuit system **2** mounted with the fixing member **1** and having a magnetic gap **20**, a voice coil

3 at least partially located into the magnetic gap **20**, a first vibration system **4** connected with the voice coil **3**, a second vibration system **5** connected with the voice coil **3** and arranged surrounding the first vibration system **4**, a printed circuit board **6** powering for the voice coil **3**, a welding piece **7**, a supporting member **8** connecting the first vibration system **4** and the second vibration system **5**, and a first housing **9** covered above the first vibration system **4**. The second vibration system **5** locates outside the first vibration system **4** and arranged coaxially with the first vibration system **4**. In present embodiment, the coaxial speaker **100** is a rectangle, and in other embodiments, the coaxial speaker **100** may also be circular or other shapes.

The fixing member **1** includes a frame **11** secured to the magnetic circuit system **2**, a supporting ring **12** secured to the second vibration system **5**, and a second housing **13** mounted with the supporting ring **12**. The frame **11** is a rectangular annular structure, and the four corners of the frame **11** are provided with a supporting leg **111**, and an accommodating portion **112** is formed between each adjacent supporting leg **111**. The supporting ring **12** is annular and sandwiched between the frame **11** and the second housing **13**.

The magnetic circuit system **2** includes a magnetic bowl **21**, a first main magnet **22** disposed on the magnetic bowl **21**, a pair of second main magnets **23** stacked on the first main magnet **22**, a plurality of first auxiliary magnets **24** spaced apart from the first main magnet **22**, a plurality of second auxiliary magnets **25** stacked on the first auxiliary magnets **24**, a main pole plate **26** sandwiched between the first main magnet **22** and the second main magnets **23**, and an auxiliary pole plate **27** sandwiched between the first auxiliary magnets **24** and the second auxiliary magnets **25**. The first main magnet **22**, the main pole plate **26**, and the second main magnets **23** are disposed stacked and form the magnetic gap **20** with the stacked structure of the first auxiliary magnets **24**, the auxiliary pole plate **27**, and the second auxiliary magnets **25**. The magnetic bowl **21** includes a carrying part **211** carrying the magnets and an extending part **212** bending and extending from the carrying part **211**. The extending part **212** is received in the accommodating portion **112** of the frame **1** in a direction of a short axis of the coaxial speaker **100**. The first auxiliary magnets **24** are received in the accommodating portion **112** of the frame **1** in a direction of a long axis of the coaxial speaker **100**.

The first main magnet **22**, the second main magnets **23**, the first auxiliary magnets **24**, and the second auxiliary magnets **25** are magnetized in a vibration direction of the coaxial speaker **100**. And the first main magnet **22** and the second main magnets **23** are set opposite each other with a same pole, that is, the first main magnet **22** and the second main magnets **23** are magnetized in opposite directions. The first auxiliary magnets **24** and the second auxiliary magnets **25** are also set opposite each other with a same pole. Furthermore, the first main magnet **22** and the first auxiliary magnets **24** are magnetized in opposite directions. In this embodiment, the number of the second main magnet **23** is two, and both of the two second magnets are stacked on the first main magnet **22**. An avoiding portion **28** is formed between the two second main magnets **23**. The auxiliary pole plate **27** has a ring structure.

The first vibration system **4** includes a first diaphragm **41** and a first support **42** connecting the first diaphragm **41** and the voice coil **3**. The first diaphragm **41** is used for generating treble sound. The sound direction of the first diaphragm **41** is upward along the vibration direction. The first diaphragm **41** is an integrated structure, and the first diaphragm **41** includes a dome portion **411**, a suspension portion **412** surrounding the dome portion **411**, and a fixation portion **413** surrounding the suspension portion **412**. The first support **42** includes a first fixing portion **421** with an annular shape connecting to the voice coil **3**, a connecting portion **422** extending inwardly horizontally from the first fixing portion **421**, and a second fixing portion **423** extending from the connecting portion **422** toward the first diaphragm **41** and connected to the first diaphragm **41**. The second fixing portion **423** is a hollow tubular structure. The connecting portion **422** connects the first fixing portion **421** and a second fixing portion **423** along a short axis direction of the coaxial speaker **100**. The avoiding portion **28** formed between the two second main magnets **23** provide the avoiding space for the connecting portion **422**.

The second vibration system **5** includes a second diaphragm **51** and a second support **52** connecting the second diaphragm **51** and the voice coil **3**. The second diaphragm **51** is used for generating bass sound. The sound direction of the second diaphragm **51** is upward along the vibration direction. The second diaphragm **51** is closer to the magnetic circuit system **2** than the first diaphragm **41**. The second diaphragm **51** is a separate structure, and the second diaphragm **51** includes a first suspension **511** with an annular hollow shape, a first dome **512** surrounding the first suspension **511**, and a second suspension **513** partially surrounding the first dome **512**. The first suspension **511** includes a first attaching portion **5111** attached fixed to the supporting member **8**. The second suspension **513** includes a second attaching portion **5131** attached fixed to the supporting ring **12**. An outer end portion of the first suspension **511** is abutted against an inner end portion of the second suspension **513**. The first dome **512** is attached to and under the first suspension **511** and the second suspension **513**. The second fixing portion **423** of the first support **42** passes through the first suspension **511** and connects to the first diaphragm **41**. The second support **52** has an annular shape, and one end of the second support **52** is connected with the first dome **512**, and the other end of the second support **52** is connected to with the voice coil **3**. In detail, the end of the second support **52** distal to the second diaphragm **51** is connected with the first fixing portion **421** of the first support **42**, so that the first fixing portion **421** is sandwiched between the second support **52** and the voice coil **3**.

The second housing **13** covers the second diaphragm **51** and is spaced apart from the supporting member **8** to form a second sound hole **131** for sounding the bass sound generated by the second diaphragm **51**.

In this embodiment, the number of the printed circuit board **6** is two, and each printed circuit board **6** is arranged in the direction of the short axis of the coaxial speaker **100**. One end of the printed circuit board **6** is secured to the frame **11**, and the other end of the printed circuit board **6** elastically supports the voice coil **3**.

The welding piece **7** is injected into the frame **11** to electrically connect the printed circuit board **6** with an external circuit.

The supporting member **8** connects the first diaphragm **41** to the second diaphragm **51**. The supporting member **8** includes a first portion **81** located out of the receiving room **10** of the fixing member **1** and a second portion **82** located in the receiving room **10**. The first portion **82** is connected with the first diaphragm **41**, and the second portion **82** is connected with the second diaphragm **51**. A size of the second portion **82** is smaller than the size of the first portion

81, and the second portion **82** extends to fix with the second main magnets **23** of the magnetic circuit system **2**.

The first housing **9** covers above the first diaphragm **41** and is mounted with the fixation portion **413** of the first diaphragm **41**. A first sound hole **91** is provided on the first housing **9** for sounding the treble sound generated by the first diaphragm **41**.

The treble and the bass units of the present invention share the magnetic circuit system, and at the same time, they share the voice coil to drive the bass support (named the second support) and the treble support (named the first support) to realize the driving of the diaphragm system (named the first diaphragm and the second diaphragm) simultaneously. The FO of the treble sound can be controlled between 2 kHz~6 kHz, and the Qts>2.

The coaxial speaker of the present invention can be applied to wearable products, such as watches, and there are two application modes: the treble unit realizes the remote SOS function, and the treble and bass unit realize the voice calls and music playback.

Comparing with the related art, in the coaxial speaker of present invention, the coaxial speaker including a fixing member having a receiving room, a magnetic circuit system mounted with the fixing member and having a magnetic gap, a voice coil at least partially located into the magnetic gap, a first vibration system connected with the voice coil, and a second vibration system connected with the voice coil and arranged surrounding the first vibration system. The second vibration system locates coaxially with the first vibration system. The first vibration system includes a first diaphragm and a first support connecting the first diaphragm and the voice coil. The second vibration system includes a second diaphragm and a second support connecting the second diaphragm and the voice coil. The first diaphragm is used for generating treble sound. The second diaphragm is used for generating bass sound.

The coaxial speaker of the present invention arranges an annular bass vibration system for providing bass sound, a treble vibration system disposed surrounded by the bass vibration system for providing treble sound. Bass sound and treble sound form a full-range speaker, providing high-quality sound. Moreover, the assembly process of the coaxial speaker is simplified by sharing voice coil, which improves assembly efficiency and reduces costs. The two diaphragms share the voice coil and magnetic circuit system to further reduce costs and improve acoustic performance.

It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiments have been set forth in the foregoing description, together with details of the structures and functions of the embodiment, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

What is claimed is:

1. A coaxial speaker, comprising:

a fixing member having a receiving room;

a magnetic circuit system mounted with the fixing member and having a magnetic gap;

a voice coil at least partially located into the magnetic gap;

a first vibration system connected with the voice coil, the first vibration system comprising:

a first diaphragm, the first diaphragm used for generating treble sound; and

a first support connecting the first diaphragm and the voice coil; and

a second vibration system connected with the voice coil and arranged surrounding the first vibration system, the second vibration system located coaxially with the first vibration system, the second vibration system comprising:

a second diaphragm, the second diaphragm used for generating bass sound, the second diaphragm being closer to the magnetic circuit system than the first diaphragm; and

a second support connecting the second diaphragm and the voice coil;

a supporting member connecting the first diaphragm to the second diaphragm, the supporting member comprising a first portion located out of the receiving room of the fixing member and a second portion located in the receiving room, the first portion connected with the first diaphragm, the second portion connected with the second diaphragm, a size of the second portion smaller than the size of the first portion.

2. The coaxial speaker as described in claim **1**, wherein the first support comprises a first fixing portion with an annular shape connecting to the voice coil, a connecting portion extending inwardly from the first fixing portion, and a second fixing portion extending from the connecting portion toward the first diaphragm and connected to the first diaphragm.

3. The coaxial speaker as described in claim **2**, wherein the second diaphragm comprises a first suspension with an annular hollow shape, a first dome surrounding the first suspension, and a second suspension partially surrounding the first dome, the second fixing portion of the first support passing through the first suspension and connected to the first diaphragm.

4. The coaxial speaker as described in claim **3**, wherein the second support has an annular shape, one end of the second support connected with the first dome, and the other end of the second support connected to with the voice coil.

5. The coaxial speaker as described in claim **4**, wherein one end of the second support distal to the second diaphragm is connected to the first fixing portion of the first support so that the first fixing portion is sandwiched between the second support and the voice coil.

6. The coaxial speaker as described in claim **1**, wherein the coaxial speaker further comprises a first housing covered above the first diaphragm, a first sound hole is provided on the first housing for sounding the treble sound generated by the first diaphragm.

7. The coaxial speaker as described in claim **1**, wherein the fixing member comprises a frame secured to the magnetic circuit system, a supporting ring secured to the second diaphragm, and a second housing mounted with the supporting ring, the second housing covering the second diaphragm and spaced apart from the supporting member to form a second sound hole for sounding the bass sound generated by the second diaphragm.

8. The coaxial speaker as described in claim **1**, wherein the second portion of the supporting member extends to fix with the magnetic circuit system.

* * * * *